Ballari Institute Of Technology & Management
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Department of Computer Science Engineering-Data Science

A Mini-Project Presentation on

"Digital Health Record Management System for migrant workers "

Presented by

RANJITA M	3BR23CD074
S CHINNI RENUKA	3BR23CD078
S GAYATHRI	3BR23CD080
TARUNASHREE	3BR23CD082

Under the Guidance of

Dr. Jagadish R.M
Project Coordinator
Dept. Of CSE-DS

Mrs .V P Anushya
Project Guide
Dept.of CSE-DS

Dr. Aradhana D
Head of the Department
Dept. of CSE-DS

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Abstract

- The *Digital Health Record Management System for Migrant Workers* is designed to create a centralized and accessible platform for storing, managing, and retrieving the medical records of migrant workers. By digitizing health records, quick access to medical data for both patients and healthcare providers, and secure data storage using authentication mechanisms. The system integrates modules for registration, record management, and hospital access, making it easier to update and share medical information efficiently.

Objectives

- To develop a secure and centralized digital platform for storing and managing migrant workers' health records.
- To enable quick access and real-time updating of medical information by both healthcare providers and workers.
- To ensure data privacy, authentication, and easy retrieval of health history across different locations.
- To support continuous healthcare and better medical decision-making through organized digital records.

Literature Survey

S. No.	Author / Year	Title / Source	Key Findings / Contributions	Relevance to Project
1.	Saleh S. et al., 2020	Sijilli: A Cloud- Based Electronic Health Record for Refugees	Proposed a portable and secure EHR platform using cloud storage and QR access for displaced populations.	Demonstrates feasibility of cloud-based health records for mobile populations like migrant workers
2.	Ndlovu K. et al., 2021	Interoperability Challenges in Linking mHealth Apps to the Record Systems	Highlighted technical issues in connecting mobile apps with hospital EHRs using HL7/FHIR Standards.	Helps designdatasharing features and integration between migrant health apps and hospitals.
3.	Tensen P. et al., 2024	Electronic Personal Health Records for Mobile Populations – Feasibility Study	Found user trust, language, and connectivity as key adoption challenges for digital health tools.	Guides user interface design and offline accessibility for migrant worker systems.

Problem Statement

- Migrant workers face missing, scattered, and inaccessible health records, leading to treatment delays and repeated tests. We provide a centralized digital system to ensure continuous, secure, and accessible healthcare..

Scope of the Project

- Provides a centralized digital platform for storing migrant workers' health records.
- Enables easy access to medical history for doctors and workers.
- Ensures timely medical interventions through notifications and reminders.
- Can be extended for mobile access, government portal integration, and health analytics.

Software/Hardware requirements

- **Software Requirements:**

- Operating System: Windows 11
- Frontend: React (with JavaScript), HTML
- Backend: Node JavaScript
- Database: MySQL
- Development Tools: VS Code ,Command prompt
- Other Tools: Browser (Chrome/Edge), ER Diagram tool (Lucidchart)

- **Hardware Requirements**
- Processor: Intel i3 or above
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: Minimum 100 GB Hard Disk
- Display: 1024x768 resolution or higher
- Network: Internet connection for online record access and storage

Functional and Non-functional requirements

Functional Requirements

1. User Registration and Authentication:

- Migrant workers, doctors, and administrators should be able to register and log in securely.

2. Health Record Management:

- Allows healthcare staff to create, update, and retrieve patient (migrant worker) medical records.

3. Doctor Access and Consultation:

- Doctors can view a worker's medical history and update diagnoses, prescriptions, or test results.

Non-functional requirements

1. Security:

- All sensitive health data must be encrypted and accessible only to authorized users.

2. Usability:

- Interface should be user-friendly and accessible even for low-literate users.

3. Scalability:

- Should handle large numbers of migrant workers across different regions.

4. Performance:

- System must retrieve health records within 2–3 seconds.

Description of Modules

1. **User (Migrant Worker) Module:**

- Enables workers to register, view their personal details, and access their health history.

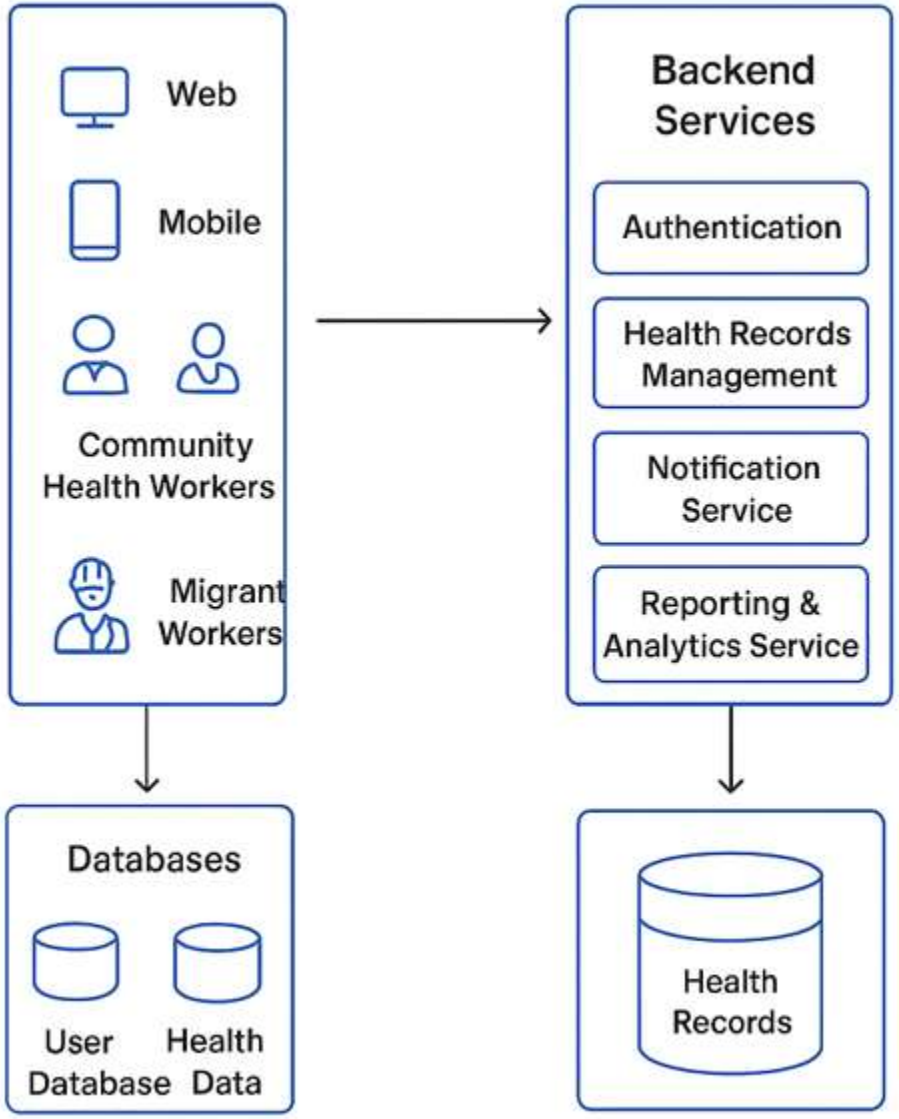
2. **Doctor Module:**

- Doctors can view medical history, add prescriptions, lab results, and recommendations.

3. **Health Record Module:**

- Stores all patient medical data — including allergies, previous treatments, vaccinations, and medical reports.

DESIGN: Digital health record system architecture diagram



Implementation

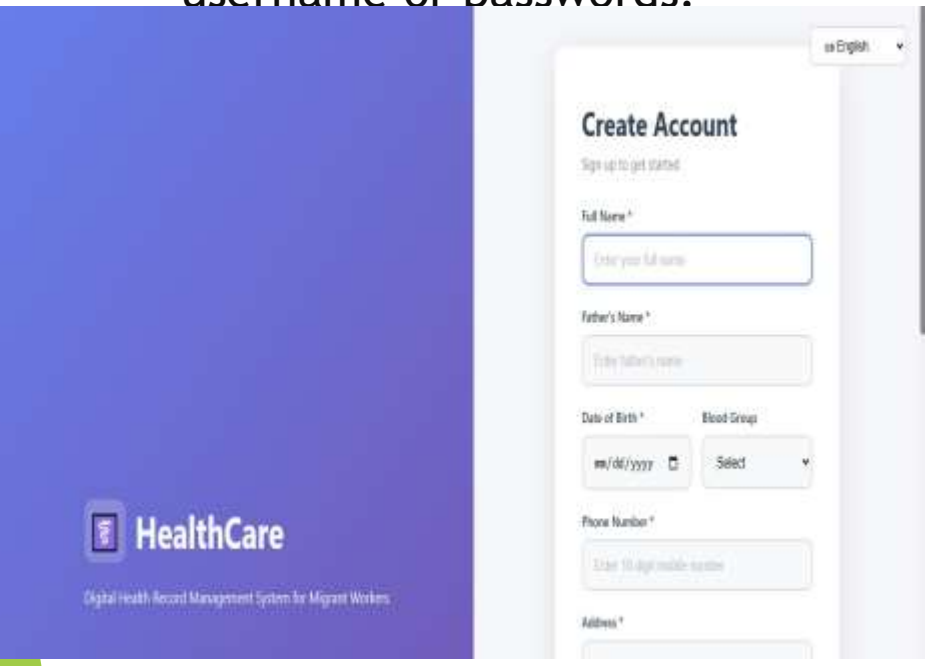
1. The project follows a clean folder structure where all main logic, components, and styles are kept inside the src folder.
2. We implemented the project using **React with TypeScript**, building the entire user interface using .
3. A main **HTML file** is used as the entry point where the React app gets loaded.
4. We used **Vite** for fast development and optimized builds through its configuration file.
5. **CSS** is integrated to style the website using utility classes for a modern, responsive UI.
6. Several **JSON configuration files** (package.json, tsconfig.json, etc.) are used to manage dependencies, scripts, and TypeScript settings.
7. A **Netlify deployment configuration** is included for easy hosting and continuous deployment.

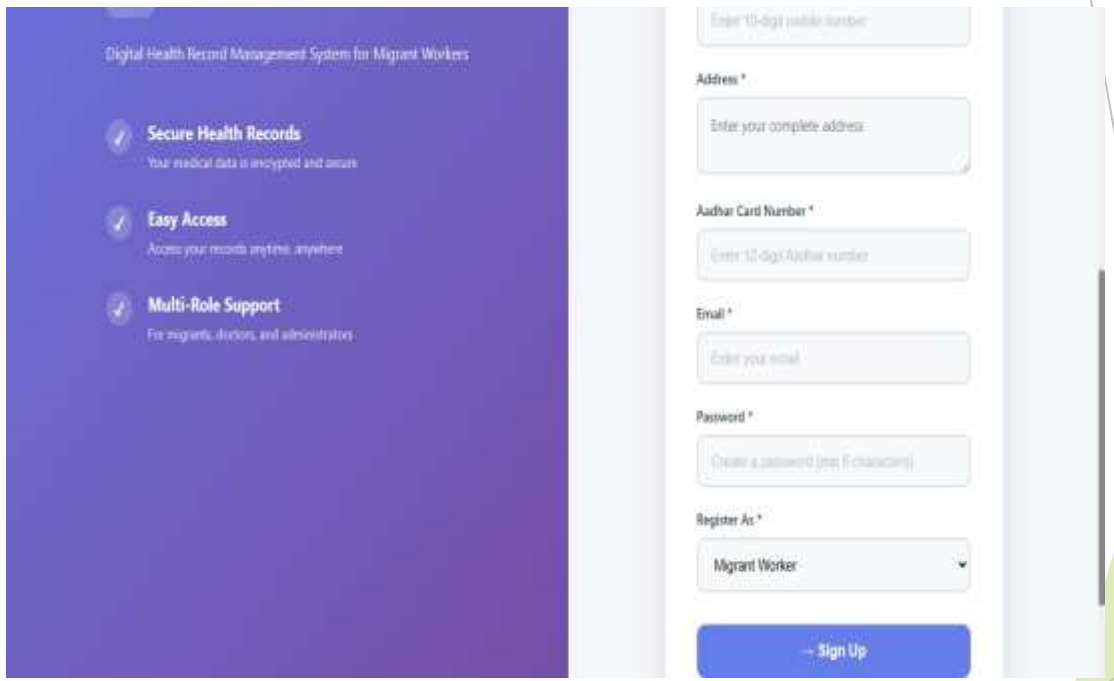
Testing

1. **Functional Testing:**
   All core features (forms, buttons, navigation, user actions) were tested to ensure they work correctly in the browser.
2. **Component Testing:**
   Individual React components were checked to ensure proper rendering and expected behavior.
3. **Responsive Testing:**
   The UI was tested across multiple screen sizes (mobile, tablet, desktop) to confirm Tailwind-based responsiveness.
4. **Browser Compatibility Testing:**
   The application was tested on major browsers like Chrome, Edge, and Firefox to ensure consistent performance and layout.
5. **Build Testing:**
   A production build was generated using Vite (npm run build) to verify that the project compiles without errors.
6. **Deployment Testing:**
   After deploying to Netlify, the live site was tested to ensure pages load correctly and all features function as expected.

Result

- Sign-up/sign-in-page
- The image shows a web based portal page for a migrant health with fields or username or passwords.





Digital Health Record System

GB English

S Gayathri (migrant_worker)

Logout

Welcome, S Gayathri

Application ID: MWCFC00012

Total Appointments

1

Health Records

0

Prescriptions

0

Appointments | Health Records | Prescriptions | Emergency Contacts | Documents

Digital Health Record System

GB English

Tanmay K (doctor)

Logout

Dr. Tanmay K

Doctor Portal - ID: DR0000013BDF

Total Patients

6

Total Appointments

2

Pending Appointments

1

Appointments | Patients | Add Health Record | Add Prescription

Type here to search

26°C ENG 12:13 PM

Emergent | Fullstack App | Sign in - Google Accounts | (26) YouTube | Digital Health Record Manager

localhost:5174

Digital Health Record System

English | **shree** (health_worker) | Logout

Register New Migrant

Name *

Age *

Gender *

Male

Native State *

Language

Contact

Unique ID *

Register Migrant

Registered Migrants (6)

Total Migrants

6

Health Records

7

Add Health Record

Disease *

Diagnosis Date *

mm/dd/yyyy

Treatment

Type here to search | Microsoft AI chief Mu... | ENG IN | 9:41 AM 11/21/2025

References

- World Health Organization (WHO). (2021). *Global strategy on digital health 2020–2025.* → https://www.who.int/publications/i/item/9789240020924
- United Nations. (2015). *Transforming our world: The 2030 Agenda for Sustainable Development.* → https://sdgs.un.org/2030agenda
- HL7 International. (2024). *FHIR (Fast Healthcare Interoperability Resources) Specification.* → https://hl7.org/fhir/
- Ministry of Health and Family Welfare, Government of India. (2022). *National Digital Health Mission (NDHM) Blueprint.* → https://ndhm.gov.in/

Conclusion:

- The Digital Health Record Management System greatly improves the daily lives of migrant workers by making their health information easily accessible, secure, and portable. It ensures quick medical care, reduces repeated tests, and helps doctors provide better treatment anywhere. Aligned with SDG goals, it promotes good health, equality, and sustainable communities by connecting people, clinics, and health services seamlessly.

THANK YOU